UC SAN DIEGO / ECE 251C

# 10

# Orthogonal filter banks

**Executive summary**

An orthogonal filter bank computes coordinates in an orthonormal basis of $\ell^2(\mathbb{Z})$ and preserves the energy of the input. We identify its basis vectors as even translates of two synthesis filters and prove that double-shift correlation conditions characterize both orthonormality and completeness. In the frequency domain, the low-pass condition becomes power complementarity. The scaling-function refinement equation supplies the additional gain normalization. Finally, the conjugate-mirror construction produces the remaining filters from one filter, giving exact reconstruction and complementary low-pass and high-pass responses.

## 10.1 Analysis filtering computes inner products

Perfect reconstruction guarantees that the outputs of a filter bank determine its input. To connect this algebraic property with the orthonormal wavelet expansions developed earlier, we must understand the geometry of the filter-bank coefficients. The connection comes from conjugate reversal: with the appropriate analysis filter, each retained output sample is an inner product with a translated synthesis filter. We begin by writing this observation explicitly.

For $x \in \ell^2(\mathbb{Z})$, let $g_0, g_1 \in \ell^2(\mathbb{Z})$ be synthesis filters and set

$$h_\ell[n] = g_\ell^*[-n], \qquad \ell \in \{0, 1\}. \tag{10.1}$$

The asterisk denotes complex conjugation. We allow complex-valued filters and signals; the conjugation disappears when a filter is real. This is the zero-delay convention used in Lecture 8.

Using $(h * x)[r] = \sum_k x[k]h[r-k]$, the two outputs are

$$\begin{aligned} a[n] &= (h_0 * x)[2n] = \sum_k x[k] g_0^*[k-2n], \\ d[n] &= (h_1 * x)[2n] = \sum_k x[k] g_1^*[k-2n]. \end{aligned} \tag{10.2}$$

Indeed, $h_\ell[2n-k] = g_\ell^*[-(2n-k)] = g_\ell^*[k-2n]$. This change in the order of subtraction is what turns convolution into an inner product.

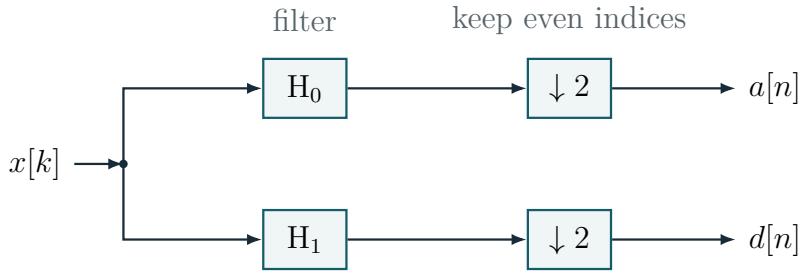



**Figure 10.1:** Filtering combines input samples. Downsampling evaluates the filtered sequence at $2n$; it does not itself perform averaging.

## 10.2 The basis vectors are even translates of two filters

Recall the inner product and its induced norm:

$$\langle x, y\rangle_{\ell^2} = \sum_{k\in\mathbb{Z}} x[k]y^*[k], \qquad \|x\|_{\ell^2}^2 = \langle x, x\rangle_{\ell^2}.$$

The *second* sequence is conjugated. Define

$$v_{\ell,n}[k] \coloneqq g_\ell[k-2n]. \tag{10.3}$$

Then (10.2) says exactly that

$$a[n] = \langle x, v_{0,n}\rangle_{\ell^2}, \qquad d[n] = \langle x, v_{1,n}\rangle_{\ell^2}.$$

The channel index $\ell$ selects one of the two generating filters, while $n$ selects its position. Retaining filtered samples at indices $2n$ corresponds to translating the generating sequence by $2n$. Thus the family relevant to the bank contains even translates of two filters, rather than all translates of one filter.

**Definition 10.1** (Orthogonal filter bank)**.** The bank with analysis filters (10.1) is *orthogonal* if

$$\mathcal{B} = \{v_{\ell,n} : \ell \in \{0,1\},\ n \in \mathbb{Z}\}$$

is an orthonormal basis of $\ell^2(\mathbb{Z})$. This requires both pairwise orthonormality and completeness: finite linear combinations of these vectors must approximate every finite-energy sequence in the $\ell^2$ norm.

**Example: Haar.** Let $g_0 = (1,1)/\sqrt{2}$ and $g_1 = (1,-1)/\sqrt{2}$, supported at $0, 1$. At each $n$, their translates occupy the pair $\{2n, 2n+1\}$. On that pair they are perpendicular unit vectors; distinct pairs do not overlap. The two vectors span all possible values on their pair, and the pairs cover $\mathbb{Z}$, so the family is also complete.

We retain the standard term *orthogonal filter bank*. When the signals and filters are complex-valued, the associated energy-preserving change of coordinates is more precisely a unitary transformation.

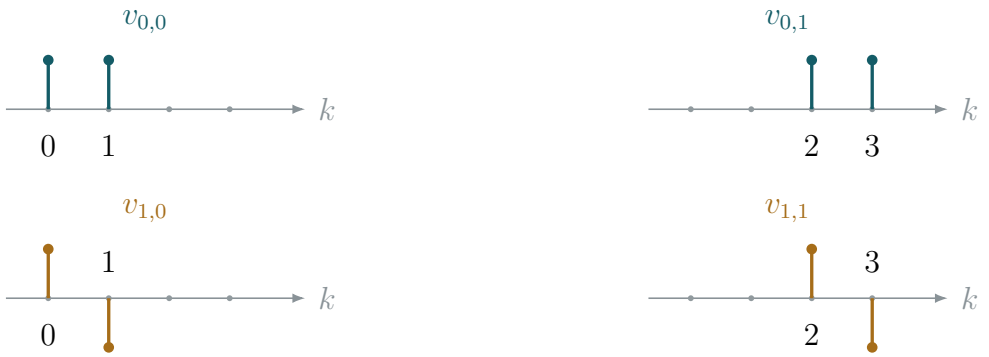



**Figure 10.2:** Two neighboring Haar pairs. Every nonzero tap has magnitude $1/\sqrt{2}$. Each pair contains an average direction and a difference direction.

## 10.3 Synthesis is the basis expansion

Suppose first that $\mathcal{B}$ is an orthonormal basis. The same coordinate rule used for Fourier series now gives

$$\begin{aligned} x[k] &= \sum_{\ell=0}^{1} \sum_{n\in\mathbb{Z}} \langle x, v_{\ell,n}\rangle_{\ell^2} v_{\ell,n}[k] \\ &= \sum_n a[n] g_0[k-2n] + \sum_n d[n] g_1[k-2n]. \end{aligned} \tag{10.4}$$

This is an expansion in the sequence space $\ell^2$, so the partial sums converge in sequence energy. The indices have distinct roles: $n$ selects a translated basis vector, whereas $k$ identifies the sample being reconstructed. Keeping these roles separate explains the argument $k-2n$ in each synthesis filter.

**Why upsampling comes before synthesis filtering.** Define the zero-inserted sequence by

$$(\uparrow 2\,a)[r] = \begin{cases} a[r/2], & r \text{ even}, \\ 0, & r \text{ odd}. \end{cases}$$

Only $r = 2n$ contributes to its convolution with $g_0$, so

$$\big(g_0 * (\uparrow 2\,a)\big)[k] = \sum_r (\uparrow 2\,a)[r] g_0[k-r] = \sum_n a[n] g_0[k-2n].$$

The second term in (10.4) is obtained in the same way from $d$.

**Energy is preserved.** For any finite set $F$ of basis indices, set $p_F = \sum_{(\ell,n)\in F} \langle x, v_{\ell,n}\rangle v_{\ell,n}$. Orthogonality makes the cross terms vanish, and gives

$$\|x - p_F\|_{\ell^2}^2 = \|x\|_{\ell^2}^2 - \sum_{(\ell,n)\in F} |\langle x, v_{\ell,n}\rangle_{\ell^2}|^2.$$

The vector $p_F$ is the orthogonal projection onto the included basis vectors, so it is the best approximation from their span. Completeness supplies finite approx-

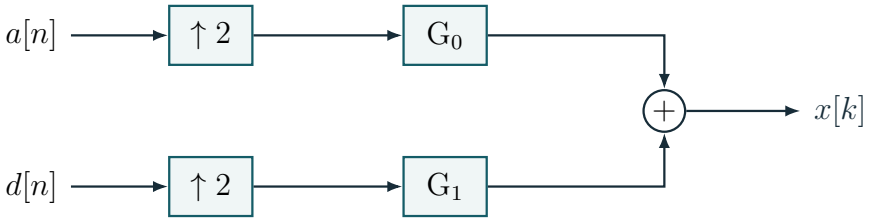



**Figure 10.3:** The synthesis bank is the basis expansion: place each coefficient on the even grid, filter with its basis generator, and add the two channels.

imations with arbitrarily small error. Once $F$ contains the vectors in such an approximation, $\|x - p_F\|_{\ell^2}$ is at least as small. Thus, as $F$ exhausts the indices,

$$\boxed{\|x\|_{\ell^2}^2 = \|a\|_{\ell^2}^2 + \|d\|_{\ell^2}^2.} \tag{10.5}$$

## 10.4 A criterion using double-shift correlations

The basis definition involves infinitely many translates, but translation invariance reduces all their pairwise inner products to correlations of the two generating filters. We use the term *double-shift correlation* because the relevant shifts occur in steps of two:

$$r_{\ell s}[k] := \sum_{i \in \mathbb{Z}} g_\ell[i] g_s^*[i - 2k]. \tag{10.6}$$

The factor two again comes from downsampling. These are correlations, not convolutions: the shifted sequence in the second factor is conjugated. For finite-energy filters, Cauchy–Schwarz ensures absolute convergence:

$$\sum_i |g_\ell[i] g_s^*[i - 2k]| \le \|g_\ell\|_{\ell^2} \|g_s\|_{\ell^2} < \infty.$$

A shift does not change a sequence's $\ell^2$ norm.

**Theorem 10.2** (Double-shift orthogonality)**.** Let $g_0, g_1 \in \ell^2(\mathbb{Z})$ and $v_{\ell,n}[i] = g_\ell[i - 2n]$. Then $\{v_{\ell,n}\}_{\ell \in \{0,1\}, n \in \mathbb{Z}}$ is an orthonormal basis of $\ell^2(\mathbb{Z})$ if and only if

$$\boxed{r_{\ell s}[k] = \delta_{\ell s} \delta[k] \quad (\ell, s \in \{0, 1\},\ k \in \mathbb{Z}).} \tag{10.7}$$

Here $\delta_{\ell s} = 1$ when the channel indices agree and zero otherwise. Explicitly,

$$r_{00}[k] = r_{11}[k] = \delta[k], \qquad r_{01}[k] = r_{10}[k] = 0.$$

*Proof.* **First establish orthonormality.** For any two channel/shift pairs,

$$\begin{aligned}\langle v_{\ell,n}, v_{s,m}\rangle_{\ell^2} &= \sum_i g_\ell[i-2n]g_s^*[i-2m] \\ &= \sum_u g_\ell[u]g_s^*[u-2(m-n)] \qquad\qquad (u = i-2n) \\ &= r_{\ell s}[m-n].\end{aligned}$$

Thus (10.7) is exactly pairwise orthonormality. It is necessary for an orthonormal basis, and it proves that our family is orthonormal when the conditions hold. We still have to prove *completeness*.

The cross conditions are not independent. Indeed, substituting $u = i + 2k$ gives

$$r_{01}^*[-k] = \sum_i g_0^*[i]g_1[i+2k] = \sum_u g_1[u]g_0^*[u-2k] = r_{10}[k].$$

It therefore suffices to check the two autocorrelations and one cross-correlation.

**How much of a unit impulse is captured?** Write $\delta_i[k] = \delta[k-i]$. Its energy is one, and

$$\langle \delta_i, v_{\ell,n}\rangle_{\ell^2} = \sum_k \delta[k-i]g_\ell^*[k-2n] = g_\ell^*[i-2n].$$

For $N \geq 0$, form the *finite* sum

$$p_{i,N} = \sum_{\ell=0}^{1}\sum_{n=-N}^{N} \langle \delta_i, v_{\ell,n}\rangle_{\ell^2} v_{\ell,n}.$$

Orthonormality shows that this finite sum is the best approximation to the impulse from the included vectors. It does not yet show that the error can be made arbitrarily small. That is precisely the completeness question we must resolve.

Put $c_{\ell,n} = \langle \delta_i, v_{\ell,n}\rangle_{\ell^2}$ for this calculation. Orthonormality gives

$$\begin{aligned}\|p_{i,N}\|_{\ell^2}^2 &= \sum_{\ell,s=0}^{1}\sum_{n,m=-N}^{N} c_{\ell,n}c_{s,m}^*\langle v_{\ell,n}, v_{s,m}\rangle_{\ell^2} \\ &= \sum_{\ell=0}^{1}\sum_{n=-N}^{N} |c_{\ell,n}|^2.\end{aligned}$$

Conjugating the coefficient in the second argument also gives

$$\langle \delta_i, p_{i,N}\rangle_{\ell^2} = \sum_{\ell=0}^{1}\sum_{n=-N}^{N} c_{\ell,n}^*\langle \delta_i, v_{\ell,n}\rangle_{\ell^2} = \sum_{\ell=0}^{1}\sum_{n=-N}^{N} |c_{\ell,n}|^2.$$

Expand the squared error:

$$\begin{aligned}\|\delta_i - p_{i,N}\|_{\ell^2}^2 &= \|\delta_i\|_{\ell^2}^2 - 2\,\mathrm{Re}\langle \delta_i, p_{i,N}\rangle_{\ell^2} + \|p_{i,N}\|_{\ell^2}^2 \\ &= 1 - \sum_{\ell=0}^{1}\sum_{n=-N}^{N} |g_\ell[i-2n]|^2.\end{aligned}$$

Since a squared norm is nonnegative, the finite sum cannot exceed one. Increasing $N$ adds nonnegative terms, so

$$S_i := \sum_{\ell=0}^{1}\sum_{n\in\mathbb{Z}} |g_\ell[i-2n]|^2 \le 1. \tag{10.8}$$

This is Bessel's inequality for the unit impulse, derived from the finite projection error. The quantity $S_i$ measures the total coefficient energy associated with that impulse. If $S_i = 1$, the error formula shows that the impulse is in the closed span of the filter translates.

**The even and odd impulses leave no missing energy.** For $i = 0$, the indices $-2n$ cover all even integers. For $i = 1$, the indices $1 - 2n$ cover all odd integers. Hence

$$S_0 = \sum_{\ell=0}^{1}\sum_{k\ \text{even}} |g_\ell[k]|^2, \qquad S_1 = \sum_{\ell=0}^{1}\sum_{k\ \text{odd}} |g_\ell[k]|^2.$$

Together they account for every tap of both filters exactly once:

$$S_0 + S_1 = \sum_{\ell=0}^{1}\sum_{k} |g_\ell[k]|^2 = \|g_0\|_{\ell^2}^2 + \|g_1\|_{\ell^2}^2 = 2.$$

The last equality is (10.7) at zero shift in each channel. But $S_0 \le 1$ and $S_1 \le 1$, so necessarily $S_0 = S_1 = 1$.

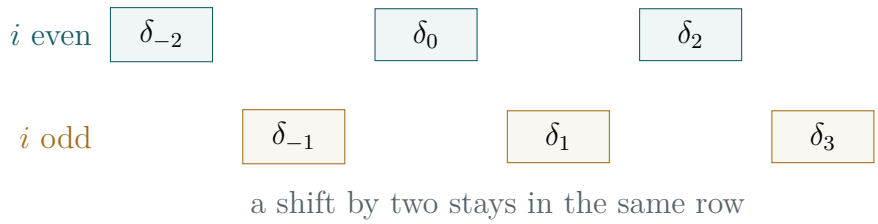



**Figure 10.4:** There are only two translation classes of unit impulses. The two unit-norm generators force the captured energy in each class to be exactly one.

For any $i = 2q + r$, with $r \in \{0, 1\}$, substitute $u = n - q$:

$$S_i = \sum_{\ell=0}^{1}\sum_{n} |g_\ell[2q + r - 2n]|^2 = \sum_{\ell=0}^{1}\sum_{u} |g_\ell[r - 2u]|^2 = S_r = 1.$$

The error formula in the preceding construction now gives $\|\delta_i - p_{i,N}\|_{\ell^2} \to 0$. Every unit impulse belongs to the closed span of our family.

Finally, any $x \in \ell^2(\mathbb{Z})$ is approximated by its finite truncations:

$$x_M = \sum_{|i| \le M} x[i]\delta_i, \qquad \|x - x_M\|_{\ell^2}^2 = \sum_{|i|>M} |x[i]|^2 \longrightarrow 0.$$

Each $x_M$ lies in that closed span, and therefore so does $x$. This proves completeness and finishes the theorem. □

The decisive step is the match between two channels and two translation classes of impulses. Orthonormality bounds the captured energy in each class by one, while the two unit-norm filters force the sum to be two. Thus neither class can have an energy deficit. This is why the correlation conditions prove completeness here, even though an arbitrary orthonormal family need not be complete.

## 10.5 Autocorrelation describes the power response

The correlation criterion is expressed in the sample domain. For design, it is useful to express the same condition in terms of the frequency response. The link is the full autocorrelation of the low-pass filter, whose Fourier transform is its squared magnitude response. Separating the even correlations will then recover the double-shift condition. From this point onward the filters are FIR, so the following rearrangements involve only finite sums of taps.

To distinguish the full autocorrelation from its even samples, write

$$r[k] := \sum_n g_0[n] g_0^*[n-k], \qquad r_{00}[m] = r[2m]. \tag{10.9}$$

The low-pass part of (10.7) therefore requires $r[2m] = \delta[m]$. It does *not* require $r[k] = 0$ at odd shifts.

**Expand the squared magnitude.** With the Fourier notation from Lecture 3,

$$\widehat{g}_0(\omega) = G_0(\mathrm{e}^{\mathrm{j}\omega}) = \sum_n g_0[n]\mathrm{e}^{-\mathrm{j}\omega n}.$$

Complex conjugation changes both the coefficient and the sign of the exponential. Consequently,

$$\begin{aligned}|\hat{g}_0(\omega)|^2 &= \hat{g}_0(\omega)\hat{g}_0(\omega)^* \\ &= \left(\sum_n g_0[n]\mathrm{e}^{-\mathrm{j}\omega n}\right)\left(\sum_m g_0^*[m]\mathrm{e}^{\mathrm{j}\omega m}\right) \\ &= \sum_{n,m} g_0[n]g_0^*[m]\mathrm{e}^{-\mathrm{j}\omega(n-m)} \\ &= \sum_k \left(\sum_n g_0[n]g_0^*[n-k]\right)\mathrm{e}^{-\mathrm{j}\omega k} \qquad (k = n-m) \\ &= \sum_k r[k]\mathrm{e}^{-\mathrm{j}\omega k}.\end{aligned}$$

Thus the *power response* $|\hat{g}_0|^2$ is the DTFT of the full autocorrelation. The inverse DTFT recovers $r[k]$ with the opposite exponential sign:

$$r[k] = \frac{1}{2\pi}\int_{-\pi}^{\pi} |\hat{g}_0(\omega)|^2 \mathrm{e}^{\mathrm{j}k\omega}\,\mathrm{d}\omega.$$

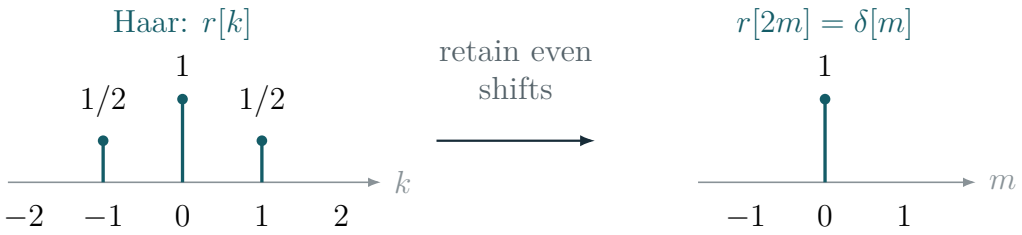



**Figure 10.5:** The Haar low-pass filter overlaps its one-sample translate, giving $r[\pm 1] = 1/2$. Only the even shifts are required to be orthogonal.

**A half-period shift selects the even correlations.** Since $\mathrm{e}^{-\mathrm{j}\pi k} = (-1)^k$,

$$\begin{aligned}|\hat{g}_0(\omega)|^2 + |\hat{g}_0(\omega+\pi)|^2 &= \sum_k r[k]\big(1 + (-1)^k\big)\mathrm{e}^{-\mathrm{j}\omega k} \\ &= 2\sum_m r[2m]\mathrm{e}^{-\mathrm{j}2m\omega}.\end{aligned}$$

The factor in parentheses is two for even $k$ and zero for odd $k$. If $r[2m] = \delta[m]$, the last sum is the constant two.

Conversely, suppose the sum of the two power responses is two for all $\omega$. Divide the preceding identity by two, multiply by $\mathrm{e}^{\mathrm{j}2q\omega}$, and integrate:

$$\begin{aligned}\frac{1}{2\pi}\int_{-\pi}^{\pi}\mathrm{e}^{\mathrm{j}2q\omega}\,\mathrm{d}\omega &= \sum_m r[2m]\,\frac{1}{2\pi}\int_{-\pi}^{\pi}\mathrm{e}^{\mathrm{j}2(q-m)\omega}\,\mathrm{d}\omega \\ \delta[q] &= \sum_m r[2m]\delta[q-m] = r[2q].\end{aligned}$$

We used the orthogonality of the integer complex exponentials, exactly as in Fourier series. This proves both directions:

$$\boxed{\begin{aligned} \sum_n g_0[n] g_0^*[n-2m] &= \delta[m] \quad \text{for every } m \\ &\Updownarrow \\ |\widehat{g}_0(\omega)|^2 + |\widehat{g}_0(\omega+\pi)|^2 &= 2 \quad \text{for every } \omega. \end{aligned}} \tag{10.10}$$

This equivalence is the *power-complementarity condition*. It constrains the squared magnitudes, which represent the response to signal energy. The magnitudes themselves need not add to a constant.

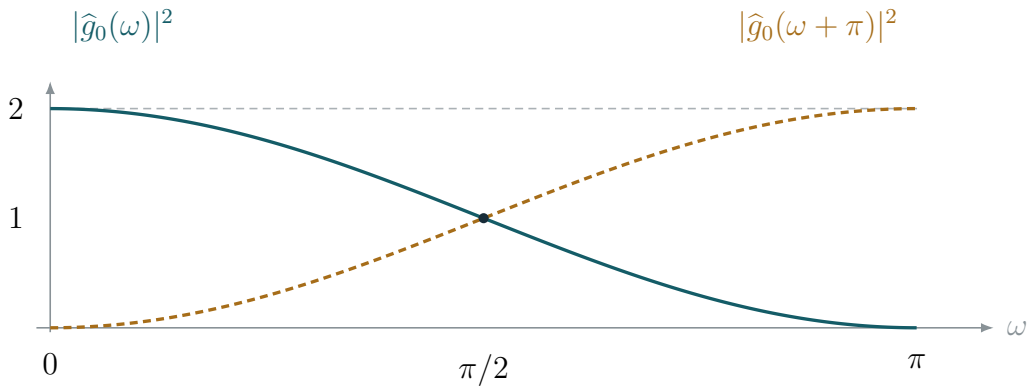



**Figure 10.6:** For Haar, the responses are $1+\cos\omega$ and $1-\cos\omega$. Their sum is two at every frequency. They both equal one at $\pi/2$.

**Orthogonality does not fix the zero-frequency gain.** The filter $g_0 = \delta$ has orthonormal even translates and satisfies power complementarity, but $G_0(1) = 1$. The refinement equation supplies an additional constraint.

## 10.6 Why the low-pass taps sum to $\sqrt{2}$

Orthogonality constrains the energy of the filter and its even translates. The refinement of an integrable scaling function with nonzero integral imposes an additional constraint on the sum of its taps. The value $\sqrt{2}$ is forced by the normalization of the two-scale equation, as the following calculation shows.

Recall the scaling-function two-scale equation from Lectures 6 and 7:

$$\phi(t) = \sqrt{2} \sum_n g_0[n] \phi(2t-n). \tag{10.11}$$

For the following calculation, assume $\phi \in L^1(\mathbb{R})$ and

$$A := \int_{\mathbb{R}} \phi(t)\, \mathrm{d}t \neq 0.$$

The usual normalization is $A = 1$, but only nonzero $A$ is needed here. Since $g_0$ is FIR, we may integrate its finite sum term by term:

$$A = \sqrt{2} \sum_n g_0[n] \int_{\mathbb{R}} \phi(2t - n)\, \mathrm{d}t.$$

For each fixed $n$, set $u = 2t - n$. Then $t = (u + n)/2$ and $\mathrm{d}t = \mathrm{d}u/2$, so

$$\int_{-\infty}^{\infty} \phi(2t - n)\, \mathrm{d}t = \frac{1}{2} \int_{-\infty}^{\infty} \phi(u)\, \mathrm{d}u = \frac{A}{2}.$$

It follows that

$$A = \sqrt{2} \sum_n g_0[n] \frac{A}{2} = \frac{A}{\sqrt{2}} \sum_n g_0[n].$$

Dividing by $A \neq 0$ gives

$$\boxed{G_0(1) = \hat{g}_0(0) = \sum_n g_0[n] = \sqrt{2}.} \tag{10.12}$$

The factor $\sqrt{2}$ in (10.11) normalizes the finer basis functions in energy; the factor $1/2$ comes from their compressed domain.

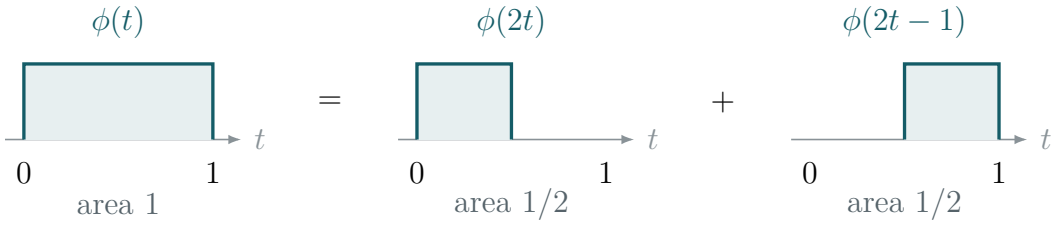



**Figure 10.7:** Haar makes the integral calculation visible: $\phi(t) = \phi(2t) + \phi(2t-1)$. Here $g_0[0] = g_0[1] = 1/\sqrt{2}$, so the two weights $\sqrt{2} g_0[n]$ equal one.

The argument derives a necessary condition from an existing refinement solution. When designing a filter first, we impose this condition so that such a solution is possible. It is not by itself an existence theorem for a scaling function or an MRA; that question is addressed in Lecture 11.

## 10.7 The even and odd tap sums are balanced

At zero frequency, (10.10) gives

$$|\hat{g}_0(0)|^2 + |\hat{g}_0(\pi)|^2 = 2.$$

Substitute (10.12): $2 + |\hat{g}_0(\pi)|^2 = 2$. Therefore

$$\hat{g}_0(\pi) = G_0(-1) = 0. \tag{10.13}$$

Here $z = 1$ corresponds to frequency zero, while $z = -1 = \mathrm{e}^{\mathrm{j}\pi}$ corresponds to frequency $\pi$.

Separate the even-indexed and odd-indexed taps:

$$E \coloneqq \sum_m g_0[2m], \qquad O \coloneqq \sum_m g_0[2m+1].$$

The $z$-transform splits as

$$G_0(z) = \sum_m g_0[2m] z^{-2m} + \sum_m g_0[2m+1] z^{-(2m+1)}.$$

At $z = 1$, both powers equal one. At $z = -1$, even powers equal one and odd powers equal minus one, also for negative exponents. Hence

$$G_0(1) = \sum_m g_0[2m] + \sum_m g_0[2m+1] = E + O = \sqrt{2},$$
$$G_0(-1) = \sum_m g_0[2m] - \sum_m g_0[2m+1] = E - O = 0.$$

Adding and subtracting these equations gives $2E = \sqrt{2}$ and $2O = \sqrt{2}$:

$$\boxed{\sum_{n \text{ even}} g_0[n] = \sum_{n \text{ odd}} g_0[n] = \frac{1}{\sqrt{2}}.} \tag{10.14}$$

**Interpretation using two simple inputs.** Let $x_c[n] = 1$ and $x_a[n] = (-1)^n$. These sequences are not in $\ell^2$, but FIR convolution is still defined. Direct substitution gives

$$(g_0 * x_c)[n] = \sum_k g_0[k] = \sqrt{2},$$
$$(g_0 * x_a)[n] = \sum_k g_0[k](-1)^{n-k} = (-1)^n \underbrace{\sum_k (-1)^k g_0[k]}_{G_0(-1)=0} = 0.$$

Thus the filter passes a constant with gain $\sqrt{2}$ and annihilates the alternating sequence. These two inputs test the responses at frequencies zero and $\pi$. They explain the low-pass endpoint behavior, but do not determine the shape of the response between those frequencies.

## 10.8 Conjugate mirror filters: one filter determines the bank

The preceding conditions suggest a more economical design. Start with one FIR filter whose even translates are orthonormal. Conjugating and reversing its taps, followed by alternating their signs, produces a second filter orthogonal to the first channel. Conjugate reversal then supplies both analysis filters. The theorem below verifies each part of this construction.

**Theorem 10.3** (Conjugate-mirror construction)**.** Suppose the complex-valued FIR filter $g_0$ is supported in $\{0, \dots, K\}$, where $K$ is odd, and satisfies

$$\sum_n g_0[n] g_0^*[n - 2m] = \delta[m], \qquad m \in \mathbb{Z}.$$

Define

$$\boxed{g_1[n] = (-1)^n g_0^*[K - n], \qquad h_\ell[n] = g_\ell^*[-n], \quad \ell = 0, 1.} \tag{10.15}$$

Then the resulting two-channel bank is orthogonal and has perfect reconstruction with zero delay.

The filter $g_1$ is the *conjugate-mirror partner* of $g_0$. With the power-complementarity condition, this is a pair of *conjugate mirror filters* (CMFs). The rule for forming the partner does not replace the orthogonality condition on the starting filter.

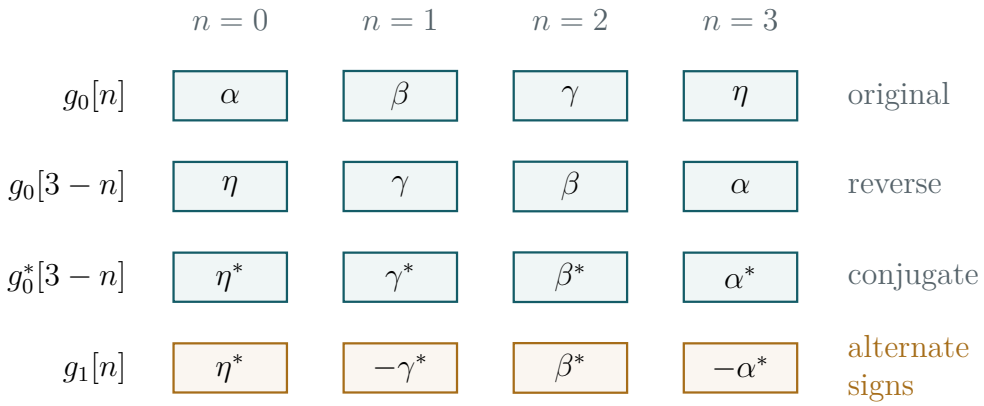



**Figure 10.8:** The construction for $K = 3$. The symbols illustrate the operations, not a particular choice of filter taps. The starting row must still satisfy double-shift orthogonality.

Both synthesis filters have support contained in $\{0, \dots, K\}$. The analysis filters have support contained in $\{-K, \dots, 0\}$. The zero-delay convention therefore allows noncausal analysis filters; it is not a claim of causal real-time processing without latency.

*Proof of the conjugate-mirror construction.* We apply theorem 10.2. The $g_0$ autocorrelation is already assumed. For the other channel, first conjugate the second factor:

$$g_1^*[n - 2k] = (-1)^{n-2k} g_0[K - n + 2k].$$

Therefore

$$\begin{aligned} r_{11}[k] &= \sum_n (-1)^n g_0^*[K-n]\,(-1)^{n-2k} g_0[K-n+2k] \\ &= \sum_n g_0^*[K-n] g_0[K-n+2k] \\ &= \sum_u g_0[u] g_0^*[u-2k] \qquad (u = K-n+2k) \\ &= r_{00}[k] = \delta[k]. \end{aligned}$$

The signs cancel because $(-1)^n(-1)^{n-2k} = 1$. In particular, the second filter has unit energy.

For the cross-correlation, the conjugation in the inner product cancels the one in the definition of $g_1$:

$$\begin{aligned} r_{01}[k] &= \sum_n g_0[n] g_1^*[n-2k] \\ &= \sum_n (-1)^n g_0[n] g_0[K+2k-n]. \end{aligned}$$

Neither factor in the last product is conjugated. Set $I = K + 2k$, which is odd, and substitute $u = I - n$:

$$\begin{aligned} r_{01}[k] &= \sum_u (-1)^{I-u} g_0[I-u] g_0[u] \\ &= -\sum_u (-1)^u g_0[u] g_0[I-u] \\ &= -r_{01}[k]. \end{aligned}$$

The second line uses $(-1)^{I-u} = -(-1)^u$. Thus $r_{01}[k] = 0$, and $r_{10}[k] = r_{01}^*[-k] = 0$ as well.

All four double-shift conditions hold. By theorem 10.2, the even translates of the two filters form an orthonormal basis, *including completeness*. Equations (10.4) and (10.5) therefore give exact zero-delay reconstruction and energy preservation. □

The cancellation uses the oddness of $K$: the paired indices $n$ and $K + 2k - n$ have opposite parity and therefore opposite modulation signs. For Haar, $K = 1$, so the same construction maps $(1, 1)/\sqrt{2}$ to $(1, -1)/\sqrt{2}$. The general formula is a direct extension of this average–difference pair.

## 10.9 Checking reconstruction in the complex $z$-domain

The basis argument already proves reconstruction for every $x \in \ell^2(\mathbb{Z})$. For the $z$-domain check, first take a finitely supported input. We verify the perfect-reconstruction identities from Lectures 8 and 9, keeping all conjugations explicit.

**Conjugate reversal.** Substituting $q = -n$ gives

$$\begin{aligned} H_\ell(z) &= \sum_n g_\ell^*[-n] z^{-n} = \sum_q g_\ell^*[q] z^q \\ &= \left[ G_\ell\left(\frac{1}{z^*}\right) \right]^*. \end{aligned}$$

The star outside the brackets conjugates the *entire value*. On the unit circle, $1/z^* = z$, so $H_\ell(\mathrm{e}^{\mathrm{j}\omega}) = G_\ell(\mathrm{e}^{\mathrm{j}\omega})^*$. For complex taps, simply replacing $z$ by $z^{-1}$ would omit the coefficient conjugation.

**The conjugate-mirror partner.** In the transform of $g_1$, put $q = K - n$. Since $K$ is odd, $(-1)^{K-q} = -(-1)^q$, and

$$\begin{aligned} G_1(z) &= \sum_n (-1)^n g_0^*[K-n] z^{-n} \\ &= \sum_q (-1)^{K-q} g_0^*[q] z^{-(K-q)} \\ &= -z^{-K} \sum_q (-1)^q g_0^*[q] z^q \\ &= -z^{-K} H_0(-z). \end{aligned}$$

For the second analysis filter, conjugating $g_1[-n]$ gives

$$h_1[n] = g_1^*[-n] = (-1)^n g_0[K+n].$$

Set $q = K + n$ in its transform:

$$\begin{aligned} H_1(z) &= \sum_n (-1)^n g_0[K+n] z^{-n} \\ &= \sum_q (-1)^{q-K} g_0[q] z^{-(q-K)} \\ &= -z^K \sum_q (-1)^q g_0[q] z^{-q} = -z^K G_0(-z). \end{aligned}$$

We have obtained

$$\boxed{G_1(z) = -z^{-K} H_0(-z), \qquad H_1(z) = -z^K G_0(-z).} \tag{10.16}$$

These are finite Laurent-polynomial identities, valid for $z \neq 0$.

**Alias cancellation.** Replacing $z$ by $-z$ in the second identity of (10.16) gives

$$H_1(-z) = -(-z)^K G_0(z) = z^K G_0(z).$$

Therefore the alias term cancels exactly:

$$\begin{aligned} &G_0(z)H_0(-z) + G_1(z)H_1(-z) \\ &\quad = G_0(z)H_0(-z) - z^{-K}H_0(-z)z^{K}G_0(z) = 0. \end{aligned}$$

**The desired term adds to two.** Define $R_0(z) = G_0(z)H_0(z)$. Expanding and grouping by $k = n - m$ gives

$$R_0(z) = \sum_{n,m} g_0[n]g_0^*[m]z^{-(n-m)} = \sum_k r[k]z^{-k}.$$

Thus

$$\begin{aligned} R_0(z) + R_0(-z) &= \sum_k r[k]\big(1 + (-1)^k\big)z^{-k} \\ &= 2\sum_m r[2m]z^{-2m} = 2\sum_m \delta[m]z^{-2m} = 2. \end{aligned}$$

Using (10.16) again,

$$\begin{aligned} G_0(z)H_0(z) + G_1(z)H_1(z) &= R_0(z) + H_0(-z)G_0(-z) \\ &= R_0(z) + R_0(-z) = 2. \end{aligned}$$

In the input–output formula from Lecture 8, these two calculations give

$$Y(z) = \frac{1}{2}(2)X(z) + \frac{1}{2}(0)X(-z) = X(z).$$

The factor two compensates for the factor $1/2$ introduced by the downsampling/upsampling identity.

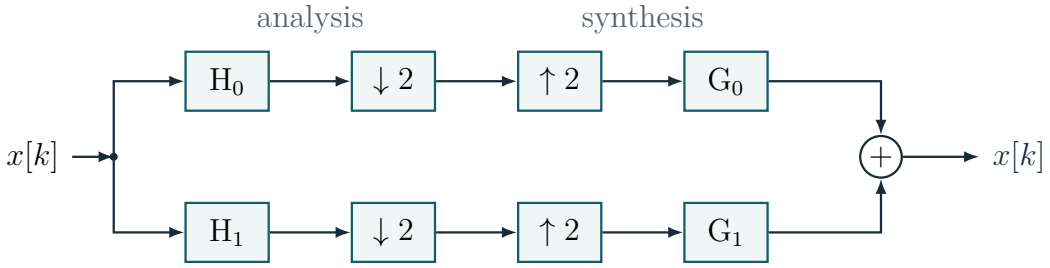



**Figure 10.9:** The conjugate-mirror bank has zero overall delay in this indexing convention. Both channels are required: their alias terms cancel only when added.

## 10.10 Mirror symmetry of the frequency responses

On the unit circle, the first identity in (10.16) reads

$$\widehat{g}_1(\omega) = -\mathrm{e}^{-\mathrm{j}K\omega}\widehat{g}_0(\omega + \pi)^*.$$

Conjugation and the phase factor do not change absolute values. Hence, even for complex-valued filters,

$$\boxed{|\hat{g}_1(\omega)| = |\hat{g}_0(\omega + \pi)|.} \tag{10.17}$$

The magnitude response is shifted by half a period. If the low-pass normalization is also imposed, then

$$G_1(1) = -G_0(-1)^* = 0, \qquad G_1(-1) = G_0(1)^* = \sqrt{2}.$$

Thus the partner rejects the constant input and passes the alternating input.

**For real taps, a shift becomes a reflection.** If $g_0$ is real,

$$\hat{g}_0(-\theta) = \sum_n g_0[n] \mathrm{e}^{\mathrm{j}\theta n} = \hat{g}_0(\theta)^*.$$

Its magnitude is therefore even. Using $2\pi$-periodicity and then this evenness,

$$\begin{aligned} |\hat{g}_1(\omega)| &= |\hat{g}_0(\omega + \pi)| = |\hat{g}_0(\omega - \pi)| \\ &= |\hat{g}_0(\pi - \omega)|. \end{aligned}$$

On $[0, \pi]$, the transformation $\omega \mapsto \pi - \omega$ reflects a point about $\pi/2$. This is the familiar mirror picture for a real low-pass/high-pass pair.

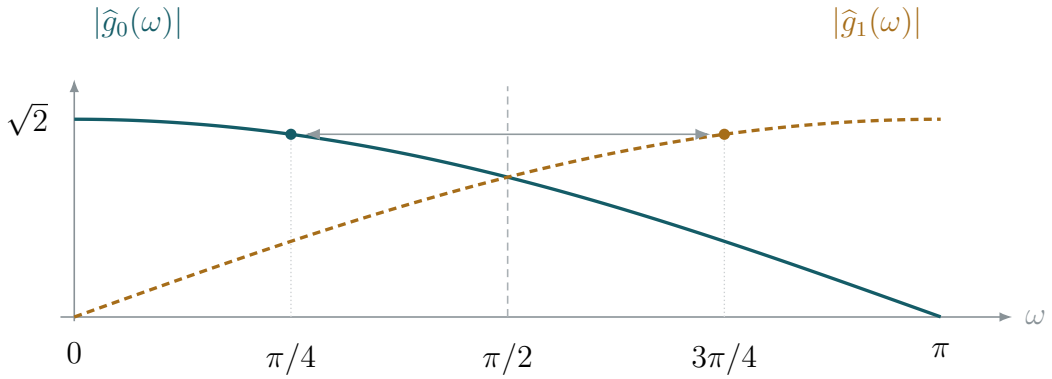



**Figure 10.10:** Haar magnitudes are $\sqrt{2}\cos(\omega/2)$ and $\sqrt{2}\sin(\omega/2)$ on $[0, \pi]$. Points at equal distances from $\pi/2$ have equal heights on opposite curves.

For complex taps, (10.17) is the general statement; the literal reflection additionally uses real-valued symmetry. The related term *quadrature mirror filter* (QMF) also occurs. Our conjugate-mirror convention is fixed by (10.15).

The construction now gives a complete orthonormal change of coordinates for sequences. To obtain an orthonormal wavelet system for functions, we must also solve the refinement equation and verify the MRA properties of its solution. Lecture 11 takes this next step, starting with the filter and constructing the scaling function through repeated refinement.